Code Source - Application Ma Calculatrice

Projet Android Native | ENSAF - Login, Menu Drawer & Calculatrice Standard/Scientifique/Graphique

1. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Ma_Calculatrice">

        <!-- Écran 1: Login (LAUNCHER) -->
        <activity
            android:name=".LoginActivity"
            android:exported="true"
            android:label="Connexion"
            android:windowSoftInputMode="adjustResize">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <!-- Écran 2: MainActivity (Calculatrice) -->
        <activity
            android:name=".MainActivity"
            android:exported="false"
            android:label="@string/app_name"
            android:windowSoftInputMode="adjustResize" />

    </application>

</manifest>
```

2. res/values/strings.xml

```
<resources>
    <string name="app_name">Ma Calculatrice</string>
    <string name="title_standard">Calculatrice Standard</string>
    <string name="title_scientific">Calculatrice Scientifique</string>
    <string name="title_devise">Convertisseur Devises</string>

    <string name="menu_standard">Standard</string>
    <string name="menu_scientific">Scientifique</string>
    <string name="menu_devise">Devise</string>
    <string name="menu_settings">Paramètres</string>
    <string name="menu_about">À propos</string>

    <string name="toast_standard">Mode Standard activé</string>
    <string name="toast_scientific">Mode Scientifique activé</string>
    <string name="toast_devise">Mode Devise (Convertisseur)</string>
    <string name="toast_settings">Paramètres de l\'application</string>
    <string name="toast_about">Ma Calculatrice v1.0 - ENSAF</string>

    <string name="navigation_drawer_open">Ouvrir le menu</string>
    <string name="navigation_drawer_close">Fermer le menu</string>
    <string name="header_title">Ma Calculatrice</string>
```

```
    <string name="header_subtitle">Calculatrice Standard &amp;amp; Scientifique</string>
    <string name="group_modes">Modes de Calcul</string>
    <string name="group_options">Options</string>
</resources>
```

3. res/values/colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFF</color>

    <!-- Palette Rose et Mauve -->
    <color name="pink_primary">#E91E63</color>
    <color name="pink_primary_dark">#C2185B</color>
    <color name="pink_accent">#FF4081</color>
    <color name="pink_light">#F8BBD0</color>

    <color name="mauve_primary">#8E24AA</color>
    <color name="mauve_dark">#4A148C</color>
    <color name="mauve_light">#E1BEE7</color>

    <color name="bg_dark_plum">#261428</color>
    <color name="bg_drawer">#331B36</color>
    <color name="bg_header">#4A1E4F</color>

    <color name="btn_digit">#4A254D</color>
    <color name="btn_operator">#E91E63</color>
    <color name="btn_control">#8E24AA</color>
    <color name="btn_scientific">#5E2B63</color>
</resources>
```

4. res/values/themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Ma_Calculatrice" parent="Theme.Material3.DayNight.NoActionBar">
        <item name="colorPrimary">@color/pink_primary</item>
        <item name="colorSecondary">@color/mauve_primary</item>
        <item name="android:statusBarColor">@color/bg_dark_plum</item>
    </style>

    <style name="Theme.Ma_Calculatrice" parent="Base.Theme.Ma_Calculatrice" />
</resources>
```

5. res/menu/main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/calc_standard"
        android:title="Standard"
        android:orderInCategory="100" />

    <item
        android:id="@+id/calc_scientific"
        android:title="Scientifique"
        android:orderInCategory="101" />

    <item
        android:id="@+id/calc_graphical"
        android:title="Graphique"
        android:orderInCategory="102" />

</menu>
```

6. res/layout/activity_login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    android:background="@color/bg_dark_plum">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Connexion"
        android:textSize="32sp"
        android:textStyle="bold"
        android:textColor="@color/pink_accent"
        android:layout_marginBottom="40dp" />

    <!-- Champ Login avec AutoCompleteTextView -->
    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        app:boxStrokeColor="@color/pink_primary"
        app:hintTextColor="@color/mauve_light"
        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">

        <AutoCompleteTextView
            android:id="@+id/etLogin"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Login"
            android:textColor="#FFFFFF"
            android:textColorHint="@color/pink_light" />
    </com.google.android.material.textfield.TextInputLayout>

    <!-- Champ Mot de passe avec TextInputEditText -->
    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="32dp"
        app:endIconMode="password_toggle"
        app:endIconTint="@color/pink_accent"
        app:boxStrokeColor="@color/pink_primary"
        app:hintTextColor="@color/mauve_light"
        style="@style/Widget.MaterialComponents.TextInputLayout.OutlinedBox">

        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Mot de passe"
            android:inputType="textPassword"
            android:textColor="#FFFFFF"
            android:textColorHint="@color/pink_light" />
    </com.google.android.material.textfield.TextInputLayout>

    <!-- Bouton de connexion -->
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="60dp"
        android:text="Se connecter"
        android:backgroundTint="@color/pink_primary"
        android:textColor="#FFFFFF"
        android:textSize="18sp" />

</LinearLayout>
```

7. res/layout/header.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="140dp"
    android:background="@color/bg_header"
    android:gravity="bottom"
    android:orientation="vertical"
    android:padding="16dp"
    android:theme="@style/ThemeOverlay.AppCompat.Dark">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Ma Calculatrice"
        android:textColor="#FFFFFF"
        android:textSize="20sp"
        android:textStyle="bold" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="4dp"
        android:text="Standard &amp;amp; Scientifique"
        android:textColor="@color/pink_light"
        android:textSize="14sp" />

</LinearLayout>
```

8. res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/drawer_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:background="@color/bg_dark_plum"
        android:padding="8dp">

        <androidx.appcompat.widget.Toolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize"
            android:background="@color/bg_dark_plum"
            app:titleTextColor="#FFFFFF" />

        <!-- Standard & Scientific Calculator Container -->
        <LinearLayout
            android:id="@+id/layoutStandardContainer"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:orientation="vertical">

            <TextView
                android:id="@+id/tvExpression"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text=""
                android:textSize="20sp"
                android:textColor="@color/mauve_light"
                android:gravity="end"
```

```
    android:padding="8dp"
    android:maxLines="2" />

<TextView
    android:id="@+id/tvDisplay"
    android:layout_width="match_parent"
    android:layout_height="100dp"
    android:text="0"
    android:textSize="52sp"
    android:textColor="#FFFFFF"
    android:gravity="end|bottom"
    android:padding="12dp"
    android:maxLines="1"
    android:autoSizeTextType="uniform" />

<LinearLayout
    android:id="@+id/layoutScientific"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:visibility="gone">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/btnSin" android:layout_width="0dp" android:layout_height="55dp" android:la
        <Button android:id="@+id/btnCos" android:layout_width="0dp" android:layout_height="55dp" android:la
        <Button android:id="@+id/btnTan" android:layout_width="0dp" android:layout_height="55dp" android:la
        <Button android:id="@+id/btnSqrt" android:layout_width="0dp" android:layout_height="55dp" android:l
        <Button android:id="@+id/btnSquare" android:layout_width="0dp" android:layout_height="55dp" androi
    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/btnLog" android:layout_width="0dp" android:layout_height="55dp" android:la
        <Button android:id="@+id/btnLn" android:layout_width="0dp" android:layout_height="55dp" android:lay
        <Button android:id="@+id/btnPi" android:layout_width="0dp" android:layout_height="55dp" android:lay
    </LinearLayout>

</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button android:id="@+id/btnClear" android:layout_width="0dp" android:layout_height="65dp" android:layo
    <Button android:id="@+id/btnBackspace" android:layout_width="0dp" android:layout_height="65dp" android:
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button android:id="@+id/btn7" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btn8" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btn9" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btnDiv" android:layout_width="0dp" android:layout_height="65dp" android:layout
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button android:id="@+id/btn4" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btn5" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btn6" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
    <Button android:id="@+id/btnMul" android:layout_width="0dp" android:layout_height="65dp" android:layout
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
```

```
            android:layout_height="wrap_content"
            android:orientation="horizontal">
            <Button android:id="@+id/btn1" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
            <Button android:id="@+id/btn2" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
            <Button android:id="@+id/btn3" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
            <Button android:id="@+id/btnSub" android:layout_width="0dp" android:layout_height="65dp" android:layout
        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">
            <Button android:id="@+id/btnDot" android:layout_width="0dp" android:layout_height="65dp" android:layout
            <Button android:id="@+id/btn0" android:layout_width="0dp" android:layout_height="65dp" android:layout_w
            <Button android:id="@+id/btnEquals" android:layout_width="0dp" android:layout_height="65dp" android:lay
            <Button android:id="@+id/btnAdd" android:layout_width="0dp" android:layout_height="65dp" android:layout
        </LinearLayout>

    </LinearLayout>

    <!-- Container for Graphical Calculator -->
    <LinearLayout
        android:id="@+id/layoutGraphical"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:visibility="gone">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:gravity="center_vertical"
            android:padding="4dp">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="f(x) = "
                android:textColor="@color/pink_light"
                android:textSize="20sp"
                android:textStyle="bold" />

            <EditText
                android:id="@+id/etFunction"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:hint="ex: sin(x), x^2, cos(x)"
                android:text="sin(x)"
                android:textColor="#FFFFFF"
                android:textColorHint="@color/pink_light"
                android:textSize="18sp" />

            <Button
                android:id="@+id/btnPlot"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Tracer"
                android:backgroundTint="@color/pink_primary"
                android:textColor="#FFFFFF" />
        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:padding="2dp">

            <Button android:id="@+id/btnPresetSin" android:layout_width="0dp" android:layout_height="48dp" android:
            <Button android:id="@+id/btnPresetCos" android:layout_width="0dp" android:layout_height="48dp" android:
            <Button android:id="@+id/btnPresetSquare" android:layout_width="0dp" android:layout_height="48dp" andro
            <Button android:id="@+id/btnPresetSqrt" android:layout_width="0dp" android:layout_height="48dp" androi
        </LinearLayout>
```

```
            <ma.ensaf.ma_calculatrice.GraphView
                android:id="@+id/graphView"
                android:layout_width="match_parent"
                android:layout_height="0dp"
                android:layout_weight="1"
                android:layout_marginTop="8dp" />

        </LinearLayout>

    </LinearLayout>

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        android:background="@color/bg_drawer"
        app:headerLayout="@layout/header"
        app:menu="@menu/main_menu"
        app:itemTextColor="#FFFFFF"
        app:itemIconTint="@color/pink_light" />

</androidx.drawerlayout.widget.DrawerLayout>
```

9. LoginActivity.java

```
package ma.ensaf.ma_calculatrice;

import android.content.Intent;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.google.android.material.textfield.TextInputEditText;

public class LoginActivity extends AppCompatActivity {

    private AutoCompleteTextView etLogin;
    private TextInputEditText etPassword;
    private Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        etLogin = findViewById(R.id.etLogin);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        // AutoCompleteTextView configuration (Course S4 Page 4-5)
        String[] suggestions = {"admin", "student", "ensaf", "user"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this, android.R.layout.simple_dropdown_item_1line, suggestions);
        etLogin.setThreshold(1);
        etLogin.setAdapter(adapter);

        // Connection intent (Course S3 Page 2-3)
        btnLogin.setOnClickListener(v -> {
            String login = etLogin.getText().toString().trim();
            String password = etPassword.getText().toString().trim();

            if (login.isEmpty() || password.isEmpty()) {
                Toast.makeText(LoginActivity.this, "Veuillez remplir tous les champs", Toast.LENGTH_SHORT).show();
            } else {
                Intent intent = new Intent(LoginActivity.this, MainActivity.class);
                startActivity(intent);
                finish();
```

```
            }
        });
    }
}
```

10. MainActivity.java

```
package ma.ensaf.ma_calculatrice;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.OnBackPressedCallback;
import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;

import com.google.android.material.navigation.NavigationView;

public class MainActivity extends AppCompatActivity implements NavigationView.OnNavigationItemSelectedListener {

    private TextView tvDisplay;
    private TextView tvExpression;
    private LinearLayout layoutScientific;
    private LinearLayout layoutStandardContainer;
    private LinearLayout layoutGraphical;

    private EditText etFunction;
    private Button btnPlot;
    private GraphView graphView;

    private DrawerLayout drawerLayout;
    private NavigationView navigationView;
    private ActionBarDrawerToggle toggle;

    private String operator = "";
    private String pendingFunction = "";
    private boolean isNewOperation = true;
    private double firstOperand = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Calculatrice Standard");
        }

        drawerLayout = findViewById(R.id.drawer_layout);
        navigationView = findViewById(R.id.nav_view);
        layoutScientific = findViewById(R.id.layoutScientific);
        layoutStandardContainer = findViewById(R.id.layoutStandardContainer);
        layoutGraphical = findViewById(R.id.layoutGraphical);

        etFunction = findViewById(R.id.etFunction);
        btnPlot = findViewById(R.id.btnPlot);
        graphView = findViewById(R.id.graphView);

        btnPlot.setOnClickListener(v -> {
```

```
    String formula = etFunction.getText().toString().trim();
    if (!formula.isEmpty()) {
        graphView.setFunctionFormula(formula);
    }
});

findViewById(R.id.btnPresetSin).setOnClickListener(v -> {
    etFunction.setText("sin(x)");
    graphView.setFunctionFormula("sin(x)");
});
findViewById(R.id.btnPresetCos).setOnClickListener(v -> {
    etFunction.setText("cos(x)");
    graphView.setFunctionFormula("cos(x)");
});
findViewById(R.id.btnPresetSquare).setOnClickListener(v -> {
    etFunction.setText("x^2");
    graphView.setFunctionFormula("x^2");
});
findViewById(R.id.btnPresetSqrt).setOnClickListener(v -> {
    etFunction.setText("sqrt(x)");
    graphView.setFunctionFormula("sqrt(x)");
});

toggle = new ActionBarDrawerToggle(
        this, drawerLayout, toolbar, R.string.navigation_drawer_open, R.string.navigation_drawer_close);
drawerLayout.addDrawerListener(toggle);
toggle.syncState();

navigationView.setNavigationItemSelectedListener(this);

getOnBackPressedDispatcher().addCallback(this, new OnBackPressedCallback(true) {
    @Override
    public void handleOnBackPressed() {
        if (drawerLayout.isDrawerOpen(GravityCompat.START)) {
            drawerLayout.closeDrawer(GravityCompat.START);
        } else {
            setEnabled(false);
            getOnBackPressedDispatcher().onBackPressed();
        }
    }
});

tvDisplay = findViewById(R.id.tvDisplay);
tvExpression = findViewById(R.id.tvExpression);

setupDigitButton(R.id.btn0, "0");
setupDigitButton(R.id.btn1, "1");
setupDigitButton(R.id.btn2, "2");
setupDigitButton(R.id.btn3, "3");
setupDigitButton(R.id.btn4, "4");
setupDigitButton(R.id.btn5, "5");
setupDigitButton(R.id.btn6, "6");
setupDigitButton(R.id.btn7, "7");
setupDigitButton(R.id.btn8, "8");
setupDigitButton(R.id.btn9, "9");

findViewById(R.id.btnDot).setOnClickListener(v -> addDot());

setupOperatorButton(R.id.btnAdd, "+");
setupOperatorButton(R.id.btnSub, "−");
setupOperatorButton(R.id.btnMul, "×");
setupOperatorButton(R.id.btnDiv, "÷");

findViewById(R.id.btnEquals).setOnClickListener(v -> calculateResult());
findViewById(R.id.btnClear).setOnClickListener(v -> clearDisplay());
findViewById(R.id.btnBackspace).setOnClickListener(v -> backspace());

findViewById(R.id.btnSin).setOnClickListener(v -> applyFunction("sin"));
findViewById(R.id.btnCos).setOnClickListener(v -> applyFunction("cos"));
findViewById(R.id.btnTan).setOnClickListener(v -> applyFunction("tan"));
findViewById(R.id.btnSqrt).setOnClickListener(v -> applyFunction("√"));
findViewById(R.id.btnSquare).setOnClickListener(v -> applyFunction("x²"));
findViewById(R.id.btnLog).setOnClickListener(v -> applyFunction("log"));
findViewById(R.id.btnLn).setOnClickListener(v -> applyFunction("ln"));
```

```
    findViewById(R.id.btnPi).setOnClickListener(v -> {
        tvDisplay.setText(String.valueOf(Math.PI));
        isNewOperation = false;
    });
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.main_menu, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(@NonNull MenuItem item) {
    int id = item.getItemId();
    return selectCalculatorMode(id) || super.onOptionsItemSelected(item);
}

@Override
public boolean onNavigationItemSelected(@NonNull MenuItem menuItem) {
    selectCalculatorMode(menuItem.getItemId());
    drawerLayout.closeDrawer(GravityCompat.START);
    return true;
}

private boolean selectCalculatorMode(int itemId) {
    if (itemId == R.id.calc_standard) {
        layoutGraphical.setVisibility(View.GONE);
        layoutStandardContainer.setVisibility(View.VISIBLE);
        layoutScientific.setVisibility(View.GONE);
        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Calculatrice Standard");
        }
        Toast.makeText(this, "Mode Standard activé", Toast.LENGTH_SHORT).show();
        return true;
    } else if (itemId == R.id.calc_scientific) {
        layoutGraphical.setVisibility(View.GONE);
        layoutStandardContainer.setVisibility(View.VISIBLE);
        layoutScientific.setVisibility(View.VISIBLE);
        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Calculatrice Scientifique");
        }
        Toast.makeText(this, "Mode Scientifique activé", Toast.LENGTH_SHORT).show();
        return true;
    } else if (itemId == R.id.calc_graphical) {
        layoutStandardContainer.setVisibility(View.GONE);
        layoutScientific.setVisibility(View.GONE);
        layoutGraphical.setVisibility(View.VISIBLE);
        if (getSupportActionBar() != null) {
            getSupportActionBar().setTitle("Calculatrice Graphique");
        }
        Toast.makeText(this, "Mode Graphique activé", Toast.LENGTH_SHORT).show();
        return true;
    }
    return false;
}

private void setupDigitButton(int buttonId, String digit) {
    findViewById(buttonId).setOnClickListener(v -> {
        if (isNewOperation) {
            tvDisplay.setText(digit);
            isNewOperation = false;
        } else {
            String currentText = tvDisplay.getText().toString();
            if (currentText.equals("0")) {
                tvDisplay.setText(digit);
            } else {
                tvDisplay.append(digit);
            }
        }
        if (!pendingFunction.isEmpty()) {
            tvExpression.setText(pendingFunction + "(" + tvDisplay.getText().toString() + ")");
        }
    });
```

```
}

private void addDot() {
    String currentText = tvDisplay.getText().toString();
    if (isNewOperation) {
        tvDisplay.setText("0.");
        isNewOperation = false;
    } else if (!currentText.contains(".")) {
        tvDisplay.append(".");
        isNewOperation = false;
    }
    if (!pendingFunction.isEmpty()) {
        tvExpression.setText(pendingFunction + "(" + tvDisplay.getText().toString() + ")");
    }
}

private void setupOperatorButton(int buttonId, String op) {
    findViewById(buttonId).setOnClickListener(v -> {
        String currentText = tvDisplay.getText().toString();
        if (currentText.isEmpty() || currentText.equals("Erreur")) return;

        if (!operator.isEmpty() && !isNewOperation) {
            calculateResult();
        }
        try {
            firstOperand = Double.parseDouble(tvDisplay.getText().toString());
            operator = op;
            tvExpression.setText(tvDisplay.getText().toString() + " " + op);
            isNewOperation = true;
        } catch (NumberFormatException e) {
            tvDisplay.setText("Erreur");
        }
    });
}

private void calculateResult() {
    String currentText = tvDisplay.getText().toString();
    if (currentText.isEmpty() || currentText.equals("Erreur")) return;

    if (!pendingFunction.isEmpty()) {
        try {
            double value = Double.parseDouble(currentText);
            double result = evaluateMathFunction(pendingFunction, value);
            if (Double.isNaN(result)) {
                tvDisplay.setText("Erreur");
                tvExpression.setText("");
                pendingFunction = "";
                isNewOperation = true;
                return;
            }
            tvExpression.setText(pendingFunction + "(" + currentText + ")");
            tvDisplay.setText(formatResult(result));
            pendingFunction = "";
            isNewOperation = true;
            return;
        } catch (Exception e) {
            tvDisplay.setText("Erreur");
            return;
        }
    }

    if (operator.isEmpty()) return;

    try {
        double secondOperand = Double.parseDouble(currentText);
        double result = 0;

        switch (operator) {
            case "+":
                result = firstOperand + secondOperand;
                break;
            case "−":
            case "-":
                result = firstOperand - secondOperand;
```

```
                break;
            case "×":
            case "*":
            case "x":
                result = firstOperand * secondOperand;
                break;
            case "÷":
            case "/":
                if (secondOperand == 0) {
                    tvDisplay.setText("Erreur");
                    tvExpression.setText("");
                    operator = "";
                    isNewOperation = true;
                    return;
                }
                result = firstOperand / secondOperand;
                break;
        }
        tvDisplay.setText(formatResult(result));
        tvExpression.setText("");
        operator = "";
        isNewOperation = true;
    } catch (Exception e) {
        tvDisplay.setText("Erreur");
    }
}

private void applyFunction(String function) {
    String currentText = tvDisplay.getText().toString();

    if (isNewOperation || currentText.equals("0")) {
        pendingFunction = function;
        tvExpression.setText(function + "(");
        isNewOperation = true;
        return;
    }

    try {
        double value = Double.parseDouble(currentText);
        double result = evaluateMathFunction(function, value);
        if (Double.isNaN(result)) {
            tvDisplay.setText("Erreur");
            return;
        }
        tvExpression.setText(function + "(" + currentText + ")");
        tvDisplay.setText(formatResult(result));
        pendingFunction = "";
        isNewOperation = true;
    } catch (Exception e) {
        tvDisplay.setText("Erreur");
    }
}

private double evaluateMathFunction(String function, double value) {
    switch (function) {
        case "sin":
            return Math.sin(Math.toRadians(value));
        case "cos":
            return Math.cos(Math.toRadians(value));
        case "tan":
            if (Math.abs(value % 180) == 90) {
                return Double.NaN;
            }
            return Math.tan(Math.toRadians(value));
        case "√":
            if (value < 0) {
                return Double.NaN;
            }
            return Math.sqrt(value);
        case "x²":
            return value * value;
        case "log":
            if (value <= 0) {
                return Double.NaN;
```

```
                }
                return Math.log10(value);
            case "ln":
                if (value <= 0) {
                    return Double.NaN;
                }
                return Math.log(value);
            default:
                return 0;
        }
    }

    private String formatResult(double result) {
        if (Math.abs(result) < 1e-12) {
            result = 0;
        } else {
            result = Math.round(result * 10000000000.0) / 10000000000.0;
        }

        if (result == (long) result) {
            return String.valueOf((long) result);
        } else {
            return String.valueOf(result);
        }
    }

    private void clearDisplay() {
        tvDisplay.setText("0");
        tvExpression.setText("");
        operator = "";
        pendingFunction = "";
        firstOperand = 0;
        isNewOperation = true;
    }

    private void backspace() {
        String currentText = tvDisplay.getText().toString();
        if (currentText.equals("Erreur")) {
            clearDisplay();
            return;
        }
        if (currentText.length() > 1) {
            tvDisplay.setText(currentText.substring(0, currentText.length() - 1));
        } else {
            tvDisplay.setText("0");
            isNewOperation = true;
        }
    }
}
```

11. GraphView.java

```
package ma.ensaf.ma_calculatrice;

import android.content.Context;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.DashPathEffect;
import android.graphics.Paint;
import android.graphics.Path;
import android.util.AttributeSet;
import android.view.View;

import androidx.annotation.Nullable;

public class GraphView extends View {

    private Paint gridPaint;
    private Paint axisPaint;
    private Paint curvePaint;
    private Paint textPaint;
```

```
private String functionFormula = "sin(x)";
private float minX = -10f;
private float maxX = 10f;
private float minY = -10f;
private float maxY = 10f;

public GraphView(Context context) {
    super(context);
    init();
}

public GraphView(Context context, @Nullable AttributeSet attrs) {
    super(context, attrs);
    init();
}

private void init() {
    gridPaint = new Paint(Paint.ANTI_ALIAS_FLAG);
    gridPaint.setColor(Color.parseColor("#4A2A4B"));
    gridPaint.setStrokeWidth(1.5f);
    gridPaint.setStyle(Paint.Style.STROKE);
    gridPaint.setPathEffect(new DashPathEffect(new float[]{8, 8}, 0));

    axisPaint = new Paint(Paint.ANTI_ALIAS_FLAG);
    axisPaint.setColor(Color.parseColor("#E1BEE7"));
    axisPaint.setStrokeWidth(3f);
    axisPaint.setStyle(Paint.Style.STROKE);

    curvePaint = new Paint(Paint.ANTI_ALIAS_FLAG);
    curvePaint.setColor(Color.parseColor("#FF4081"));
    curvePaint.setStrokeWidth(5f);
    curvePaint.setStyle(Paint.Style.STROKE);

    textPaint = new Paint(Paint.ANTI_ALIAS_FLAG);
    textPaint.setColor(Color.parseColor("#F8BBD0"));
    textPaint.setTextSize(26f);
}

public void setFunctionFormula(String formula) {
    this.functionFormula = formula;
    invalidate();
}

@Override
protected void onDraw(Canvas canvas) {
    super.onDraw(canvas);

    int width = getWidth();
    int height = getHeight();
    if (width == 0 || height == 0) return;

    canvas.drawColor(Color.parseColor("#1F0F21"));

    float originX = width / 2f;
    float originY = height / 2f;

    float scaleX = width / (maxX - minX);
    float scaleY = height / (maxY - minY);

    // Draw Grid
    for (int i = (int) minX; i <= (int) maxX; i += 2) {
        if (i == 0) continue;
        float px = originX + i * scaleX;
        canvas.drawLine(px, 0, px, height, gridPaint);
        canvas.drawText(String.valueOf(i), px - 12, originY + 30, textPaint);
    }

    for (int j = (int) minY; j <= (int) maxY; j += 2) {
        if (j == 0) continue;
        float py = originY - j * scaleY;
        canvas.drawLine(0, py, width, py, gridPaint);
        canvas.drawText(String.valueOf(j), originX + 10, py + 8, textPaint);
    }
```

```
        // Draw Axes
        canvas.drawLine(0, originY, width, originY, axisPaint);
        canvas.drawLine(originX, 0, originX, height, axisPaint);

        canvas.drawText("X", width - 30, originY - 15, textPaint);
        canvas.drawText("Y", originX + 15, 35, textPaint);

        // Draw Function Curve
        Path path = new Path();
        boolean firstPoint = true;

        for (int pixelX = 0; pixelX < width; pixelX += 2) {
            double mathX = minX + (pixelX / (float) width) * (maxX - minX);
            double mathY = evaluateFunction(functionFormula, mathX);

            if (Double.isNaN(mathY) || Double.isInfinite(mathY)) {
                firstPoint = true;
                continue;
            }

            float pixelY = (float) (originY - mathY * scaleY);

            if (pixelY < -height || pixelY > height * 2) {
                firstPoint = true;
                continue;
            }

            if (firstPoint) {
                path.moveTo(pixelX, pixelY);
                firstPoint = false;
            } else {
                path.lineTo(pixelX, pixelY);
            }
        }

        canvas.drawPath(path, curvePaint);
    }

    private double evaluateFunction(String formula, double x) {
        if (formula == null || formula.isEmpty()) return x;
        String expr = formula.toLowerCase().trim().replace(" ", "");

        try {
            if (expr.contains("sin")) return Math.sin(x);
            if (expr.contains("cos")) return Math.cos(x);
            if (expr.contains("tan")) {
                double val = Math.tan(x);
                return Math.abs(val) > 20 ? Double.NaN : val;
            }
            if (expr.contains("sqrt") || expr.contains("√")) return x < 0 ? Double.NaN : Math.sqrt(x);
            if (expr.contains("log")) return x <= 0 ? Double.NaN : Math.log10(x);
            if (expr.contains("ln")) return x <= 0 ? Double.NaN : Math.log(x);
            if (expr.contains("x^2") || expr.contains("x²") || expr.contains("x*x")) return x * x;
            if (expr.contains("x^3") || expr.contains("x³")) return x * x * x;
            if (expr.contains("1/x")) return x == 0 ? Double.NaN : 1.0 / x;

            if (expr.equals("x")) return x;
            if (expr.equals("-x")) return -x;

            return evaluateSimpleExpression(expr, x);
        } catch (Exception e) {
            return Double.NaN;
        }
    }

    private double evaluateSimpleExpression(String expr, double x) {
        expr = expr.replace("x", "(" + x + ")");
        try {
            if (expr.contains("+")) {
                String[] parts = expr.split("\\+");
                double sum = 0;
                for (String p : parts) sum += evaluateSimpleExpression(p, x);
                return sum;
            }
```

```
            if (expr.contains("-") && !expr.startsWith("-")) {
                String[] parts = expr.split("-");
                double diff = evaluateSimpleExpression(parts[0], x);
                for (int i = 1; i < parts.length; i++) diff -= evaluateSimpleExpression(parts[i], x);
                return diff;
            }
            if (expr.contains("*")) {
                String[] parts = expr.split("\\*");
                double prod = 1;
                for (String p : parts) prod *= evaluateSimpleExpression(p, x);
                return prod;
            }
            return Double.parseDouble(expr.replace("(", "").replace(")", ""));
        } catch (Exception e) {
            return x;
        }
    }
}
```